

Exercise 1: Numerical Solution of the Volume Rendering Integral

Part a)

In the equation, there are two main variables: $\tilde{C}_i$ (which is about color) and A_j (which is about opacity). When we add lighting effects like Phong shading, it changes how bright or dark a point looks based on the light. This means the color changes.

So, only $\tilde{C}_i$ will change. The opacity A_j doesn't change because it is a property of the material itself, it doesn't care about the light.

Part b)

First, we know the formula for opacity:

$$A = 1 - e^{-\kappa \cdot \Delta s}$$

We need to leave κ alone to see what it is equal to.

$$e^{-\kappa \cdot \Delta s} = 1 - A$$

Let's take the natural log (ln) of both sides:

$$-\kappa \cdot \Delta s = \ln(1 - A)$$

$$\kappa = -\frac{\ln(1 - A)}{\Delta s}$$

Now, we have a new step size called $\Delta s'$, and we want to find the new opacity called A' . The formula is similar:

$$A' = 1 - e^{-\kappa \cdot \Delta s'}$$

Now we will put the κ we just found into this formula:

$$A' = 1 - e^{-\left(-\frac{\ln(1-A)}{\Delta s}\right) \cdot \Delta s'}$$

The minus signs cancel out:

$$A' = 1 - e^{\ln(1-A) \cdot \frac{\Delta s'}{\Delta s}}$$

Using logarithm rules, we can write it like this:

$$A' = 1 - (1 - A)^{\frac{\Delta s'}{\Delta s}}$$

What happens if we increase Δs ? If we look at the first formula $A = 1 - e^{-\kappa \cdot \Delta s}$, increasing Δs makes the $e^{-\kappa \cdot \Delta s}$ part smaller (closer to zero). Because we are subtracting this smaller number from 1, the final result A becomes bigger.

So, increasing Δs will **increase** A .